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ENVIRONMENTAL PHYSIOLOGY

----- Active space --

On level lands :

Atmospheric pressure = 760 mmHg.

O_2 content : 21 %.

PO_2 = 21 % of 760 = 160 mmHg.

High Altitude Physiology

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Atmospheric pressure ↓ → ↓ PO_2 → Hypoxia.

Acclimatisation (Compensatory changes) :

Mechanism	Response
↑ ventilation (Earliest change) (D/t activation of peripheral chemoreceptors)	• Respiratory alkalosis (CO_2 washout → ↓ PCO_2, ↓ H^+). • Hypocalcemia → Tetany (↓ free form of Ca^{2+}).
Erythropoietin stimulation	Polycythemia (↑ RBC count)
↑ vascular endothelial growth factors (VEGF)	↑ vascularity (D/t ↑ angiogenesis). ↑ O_2 diffusion.
Cellular acclimatisation (↑ cytochrome oxidase activity)	↑ O_2 utilization
Release of oxygen	Right shift on oxygen dissociation curve

Clinical aspect :

Normal mechanism :

- Respiratory alkalosis → ↑ HCO_3^-, ↓ PCO_2 & ↓ H^+.
- ↓ H^+ → ↑ Ca^{2+} – albumin binding → ↓ free Ca^{2+} → Tetany.
  - H^+ & Ca^{2+} compete for binding with albumin.

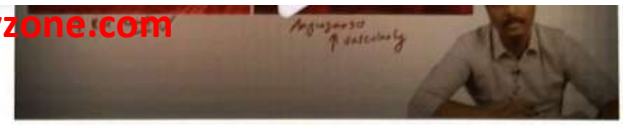
Renal compensation :

Respiratory alkalosis → ↑ HCO_3^- → Excreted via kidney → ↓ HCO_3^- → Prevents respiratory alkalosis & hence prevents tetany.

High altitude illness :

	Acute mountain sickness	Chronic mountain sickness/Monge's disease
Features	• 6–12 hrs after ascent. • C/f : Nausea, vomiting, confusion. • Complications : – High altitude pulmonary edema (HAPE). – High altitude cerebral edema (HACE).	C/f : Prolonged hypoxia (D/t ↑ erythrocytosis) ↓ Pulmonary hypertension ↓ Cor pulmonale
Rx	• Immediate descent. • O_2 therapy. • Ca^{2+} channel blocker (Nifedipine). • Dexamethasone.	• Venesection. • Acetazolamide.

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Deep Sea Physiology

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Etiology :



Divers



Military operation

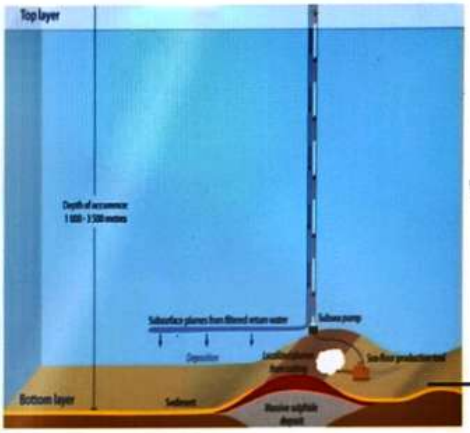


Caisson workers



Recreation activities

Mechanism :



At sea level :
Pressure = 1 atm
↓ For every 10 m
Pressure ↑ by 1 atm

High barometric pressure (gases compressed)

Applied aspect :

Caisson's disease/diver's disease/decompression sickness :

MOA : High barometric pressure $\xrightarrow{\text{quick ascent}}$ Low barometric pressure → Compressed gas (N_2) released as bubbles.

Features : Accumulation of N_2 bubbles causes,

- On crossing blood brain barrier : Euphoria (N_2 narcosis).
- In lungs : Chokes.
- In joints : Bends.
- In blood vessel : Obstruction → Air embolism → Death.

Rx :

Hyperbaric chamber
↓
Slow ascent
↓
Slow decompression (less bubble formed)



Note : Decompression sickness also seen in pilots & mountaineers.

Space Physiology

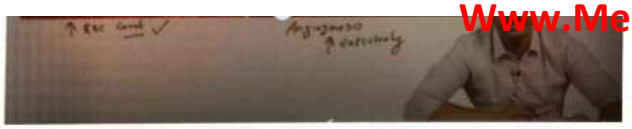
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Effects of microgravity :

- Loss of Ca^{2+} & PO_4^{3-} from bones → Prone to fractures } ↓ performance of activity
- Loss of muscle mass
- ↓↓ RBC count.

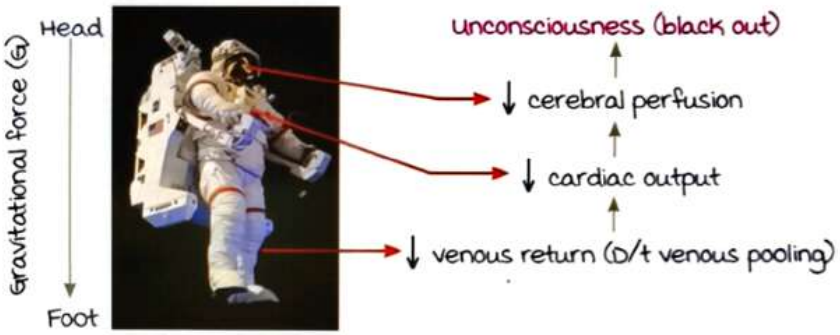
Feedback

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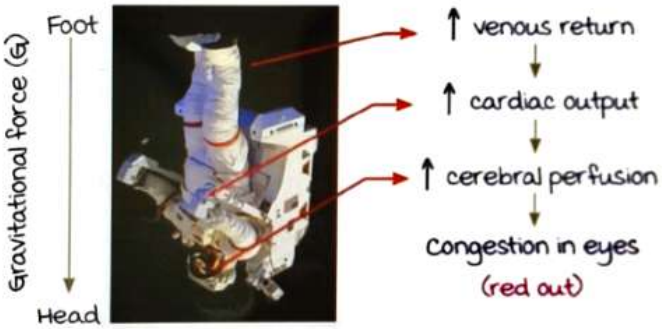


Positive G:

----- Active space -----



Negative G:



Feedback